

Project Overview

Course Name: Operating System (004310-006)

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Project Name: Costareen AI Sales Assistant Agent Using n8n Automation

1. What Problem Will This Project Solve in the Real World?

Small businesses often receive many repeated customer messages every day. Customers ask about product availability, color, price, product recommendations, and how to place an order. If a business owner or sales representative replies manually, it can take a lot of time. Customers may also receive late replies or incorrect stock information.

This project solves that real-world problem by creating an AI sales assistant for Costareen, a business that sells 3D printers, filaments, and accessories. The assistant can answer customer questions automatically in a natural sales style. It can understand customer messages, check product stock information from Google Sheets, and reply quickly.

The assistant is designed to communicate like a real sales representative named Faujia Alvi from Costareen. It can reply in English, Bangla, and Banglish, depending on the language used by the customer. This makes the conversation feel more natural for local customers.

This system helps the business by:

- Reducing manual workload for sales staff
- Giving faster replies to customers
- Checking product stock from Google Sheets
- Helping customers choose 3D printers, filaments, and accessories
- Supporting English, Bangla, and Banglish communication
- Improving customer service and sales response time
- Avoiding exact stock number sharing by only saying available or out of stock

2. Plan and Implementation Plan

The project is built using n8n as the main automation platform. The workflow connects a chat message trigger, an AI Agent, Google Gemini, conversation memory, and Google Sheets tools.

When a customer sends a message, the n8n Chat Trigger starts the workflow. The message is sent to the AI Agent. The AI Agent uses Google Gemini to understand the customer question and generate a short, polite, and helpful reply. The AI Agent can also use Google Sheets tools to check stock and sales-related information.

Main Features

- Receive customer messages through the n8n Chat Trigger

- Understand customer questions using an AI Agent node
- Generate replies using Google Gemini
- Remember recent conversation context using Simple Memory
- Check product stock information from Google Sheets
- Read sales-related data from Google Sheets
- Reply in English, Bangla, or Banglish
- Give short and natural sales responses
- Provide advice about 3D printers, filaments, and accessories
- Hide exact stock numbers and only say available or out of stock

Implementation Steps

1. Create a Google Sheet database for Costareen product and sales information.
2. Add product stock information in the `STOCK_VIEW` sheet.
3. Add sales-related information in the `SALES_ENTRY` sheet.
4. Create an n8n workflow.
5. Add the **When chat message received** trigger node.
6. Add an **AI Agent** node to process the customer message.
7. Connect the AI Agent with the **Google Gemini Chat Model**.
8. Add **Simple Memory** to remember the recent conversation context.
9. Connect the AI Agent with the Google Sheets stock lookup tool.
10. Connect the AI Agent with the Google Sheets sales entry lookup tool.
11. Write a system message that tells the assistant to behave like a Costareen sales representative.
12. Test the workflow by asking customer-style questions about 3D printers, filaments, colors, and stock availability.

Tools and Technologies

- n8n automation workflow
- n8n AI Agent node
- n8n Chat Trigger node
- Google Gemini Chat Model
- Google Sheets
- Google Sheets Tool node in n8n
- Simple Memory node
- LangChain nodes in n8n

Example Customer Message

Is the White filament available?

Example AI Reply

Yes, available! Which brand do you want?

Another Example Customer Message

Which 3D printer is good for beginners?

Example AI Reply

For beginners, an easy-to-use FDM printer is a good choice. What is your budget? I can suggest a suitable model.

Future Improvement

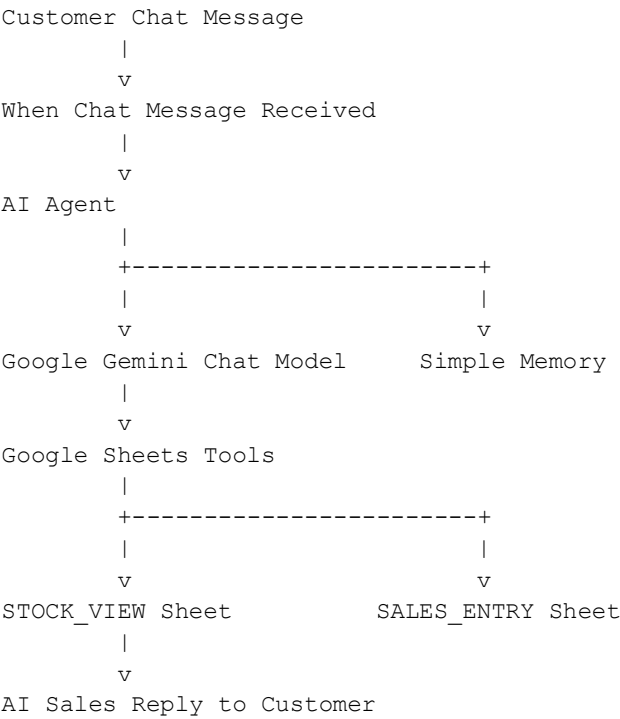
In the future, this system can be improved by connecting it with WhatsApp, Messenger, Instagram, or Telegram. It can also automatically save confirmed orders into Google Sheets, notify the business owner when a customer is ready to buy, update stock after purchase, and collect customer lead information.

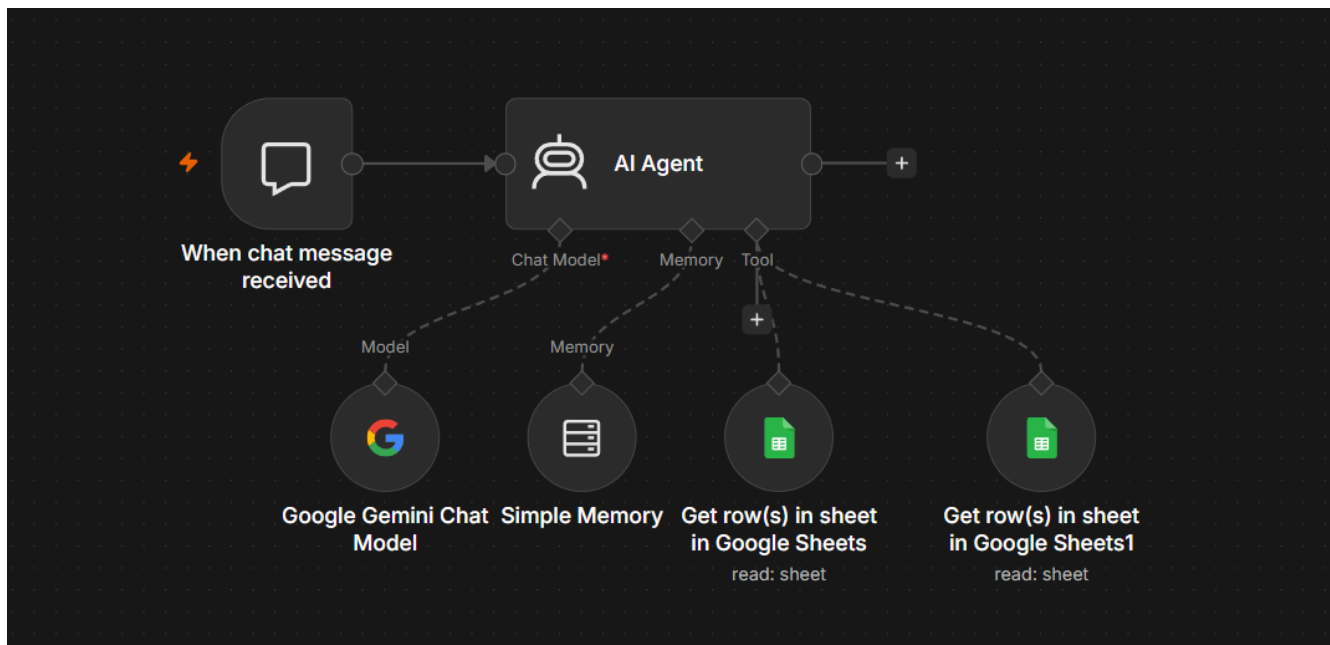
More improvements can include:

- WhatsApp Business API integration
- Facebook Messenger integration
- Automatic order entry
- Admin notification system
- Product recommendation based on customer budget
- Sales analytics dashboard
- Automatic follow-up messages

3. System Diagram

The diagram below represents the actual n8n workflow used in this project.





In this workflow, n8n works as the central automation system. It receives the customer message, sends it to the AI Agent, uses Google Gemini for response generation, checks Google Sheets for product and sales information, and produces a helpful customer reply.

Uploaded Project Files

- <https://github.com/abidurrahoman2000-sudo/Costareen-Ai-Sales-Assistant>